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PRINCIPAL ENGINEER | DEEP LEARNING DATA SCIENTIST

Highly proficient AI Scientist and Engineer adept at Reinforcement Learning, Machine Learning, and Artificial Intelligence. Known for leading initiatives in robotic learning, computer vision, face recognition and detection, object categorization, and recommendation systems. Recognized for pioneering solutions in online learning, automatic rule learning, natural language processing, knowledge representation, and energy-based algorithms. Passionate about mentoring to enhance learning and collaborating on assignments that explore and discover new technologies.

AREAS OF EXPERTISE

Artificial Intelligence | AI for Science | AI for Robotic Learning | Applied Research | AI Leadership | Mentorship | Validation Manufacturing | Collaboration | Combinatorial Optimization | Silicon Manufacturing | Foundational Models | Imitation Learning | Machine Learning Algorithms | Computational Learning | Circuit Modeling | Neuroevolution | Computer Vision | Deep Reinforcement Learning

PROFESSIONAL EXPERIENCE

Humanoid, Knoxville, Tennessee

Principal Engineer, January 2025 – Present

- Developer Lead of the Visual Language Action for the company.
- Leading advanced applied research and development in humanoid robotics, focusing on integrating AI-driven solutions to enhance robotic functionalities.
- Collaborating with cross-functional teams to design and implement machine learning algorithms for improved human-robot interactions.
- Mentoring junior engineers and scientists, fostering a culture of continuous learning and innovation.

Intel Corporation, Remote – Chandler, Arizona

Principal Engineer, April 2020 – August 2024

- Led Deep Reinforcement Learning (DRL) research in the AI Lab at Intel Labs.
- Conducted and guided applied research on the use of Large Language Models (LLM) and Reinforcement Learning (RL) for validation of silicon manufacturing, achieving potential savings of millions of dollars per month.
- Collaborated on solving constrained combinatorial optimization problems in Electronic Design Automation (EDA), resulting in a 30% speed improvement in floorplanning techniques.

- Published DRL and Imitation Learning research in AI for Science in collaboration with MILA and the University of Toronto.
- Co-authored research on Goal Conditioned DRL with Pieter Abbeel.
- Published foundational models in Language Conditioned Imitation Learning research with Heni Ben Amor at Arizona State University, recognized as a NeurIPS Spotlight Paper.
- Conducted Model-Based DRL research with Chelsea Finn at Stanford.
- Researched Imitation Learning from Perception on Robotic Surgery with Ken Goldberg.
- Conducted additional AI/ML research; see publications for details.

Senior Deep Learning Data Scientist, September 2017 – 2020

- Applied Multimodal Deep Neural Networks for retail data from external customer engagements.
- Developed Deep Learning models for recommendations in Intel's ngraph/neon front end.
- Provided internal consultancy on machine learning algorithms and tools.
- Conducted applied Deep RL research for robotics control.

Machine Learning and AI R&D, December 2013 – September 2017

- Worked on Natural Language Processing using vector spaces of meaning.
- Developed cognitive computational learning models.
- Served as a Computer Vision expert internal consultant.
- Developed and optimized ML algorithms for productization, including deployment on ASUS Zenfone2, Digital Home Space, and Real Speech Platform.

ADDITIONAL RELEVANT EXPERIENCE (2006 – 2013)

- Machine Learning, Data Science, and AI Applied Researcher: Created a humanoid robotic remote interface for the Intelligent System Group, receiving Intel Division Recognition Award in 2012.
- Internship at Intel: Explored 3D user interface prototypes for digital home applications.

EDUCATION

- Doctor of Philosophy (PhD), Computer Science, Arizona State University, Tempe, AZ
- Master of Science (MS), Computer Science, Arizona State University, Tempe, AZ

LICENSE & CERTIFICATION

- Licentiate in Computer Science, Universidad Nacional del Comahue, Neuquén, Argentina

SOFTWARE | TECHNOLOGY

- **Programming Languages:** Python | C++ | C | Java | PROLOG | LISP | Clingo | SQL | Haskell | Scala
- **Tools & Frameworks:** Cursor.AI | Microsoft Visual Studio | JAX | FLAX | PyTorch | TensorFlow | Keras | Weka | R | MATLAB | Octave | OpenGL | Git | ROS1|2
- **Other Skills:** Reinforcement Learning | Computer Vision | NLP | Data Science | Scientific Visualization

NOTABLE ASSIGNMENTS

- Keynote Speaker: CoRL Workshop (2024)
- Moderator: Senior Internal AI Conference (2023)
- Facilitator: Intel AI conference with ~4000 attendees (January 2022)
- Mentor: State-of-the-Art AI (2022); IL Circle Event on Career Growth; AI Intel Challenge Board (2022 – 2023)
- Podcaster: Co-host of "Meta Learning for Robotics" with Amir Khosrowshahi and Chelsea Finn

PUBLICATIONS (See Google Scholar for full list)

- *AI for Science*: Published multiple papers in Digital Discovery and NeurIPS in collaboration with MILA and the University of Toronto.
- *AI for EDA*: Contributions to ICCAD 2024 and IEEE - ITC 2024.
- *Robot Learning*: Research on accelerating visual sparse-reward learning, model-based adversarial imitation learning, and modular language-conditioned robot policies.
- *Foundational Model for Circuit Modeling & Control*: Work on pretraining graph neural networks and learning transferable policies.
- *Deep RL*: Developed innovative techniques such as Determinantal Point Process based neural network sampling.

PATENTS (See Google Scholar for full list)

- Maximizing Performance while Minimizing Communication in Multi-Agent Reinforcement Learning (2022)
- System and method of using neuroevolution-enhanced multi-objective optimization for mixed-precision quantization of deep neural networks (2022)
- System and method for pruning filters in deep neural networks (2022)
- Technologies for predictive monitoring of a characteristic of a system (2019)
- Technologies for physical assault detection using secure clothing and/or clothing accessories (2019)
- Dialogue system with audio watermark (2017)

REVIEWER (CURRENT)

- NeurIPS, ICML, ICLR, AAAI, CoRL.
- IEEE Robotics and Automation (2022)
- ALTA 2019: 17th Annual Workshop of The Australasian Language Technology (2019)

Google Scholar

Arxiv

- https://scholar.google.com/citations?hl=en&user=YArRsvEAAAAJ&view_op=list_works&sortby=pubdate
- <https://arxiv.org/search/cs?searchtype=author&query=Phielipp,+M>